Work at the Mdls Lab Documentation

Mobbili Leonardo

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1 Analyze the dataset

The first step I needed to do was to have a complete description of the crucial columns of the dataset that I needed to work on. This is important to understand the data and to be able to analyze the dataset. I then loaded the data into a jupyter notebook and using pandas and matplotilib I did basic statistics and plotting to try to understand the data better. After that I filtered the dataset to keep only the columns that were relevant for my analysis and created a new Excel file to store the filtered data.

1.1 Plotting and statistics

The first statistic that I did was to calculate the mean and plot the age distribution of the patients in the dataset.

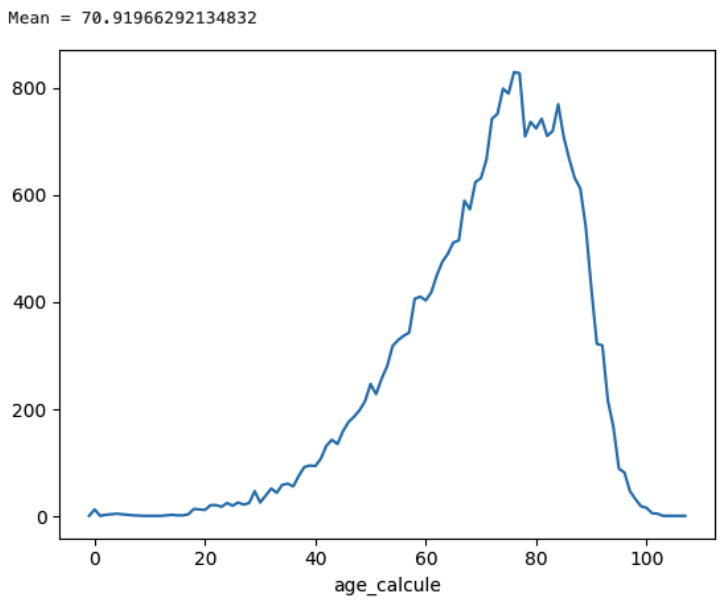

Figure 1: Age distribution plot

I then procided to calculate for each column how many missing values there were. (see attached file) .
I calculated the sex distribution of the patients in the dataset and plotted it as a pie chart.

```
Counts:
 sexe
Homme    15986
Femme    14680
Name: count, dtype: int64

Percentages:
 sexe
Homme    52.129394
Femme    47.870606
Name: count, dtype: float64
```

Figure 2: Sex distribution counts and percentages

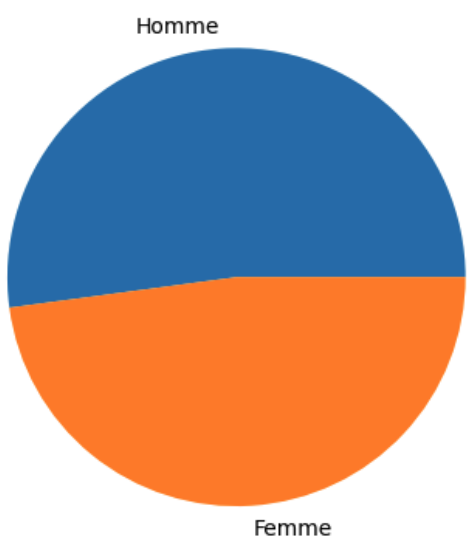

Figure 3: Sex distribution plot

The next step was to calculate the distribution of the patients between the different centers involved in the study. The full distribution is shown in the attached file. (see attached file) .
I also plotted the distribution of the patients between the different centers as a pie chart leaving out the centers with less than 100 patients.

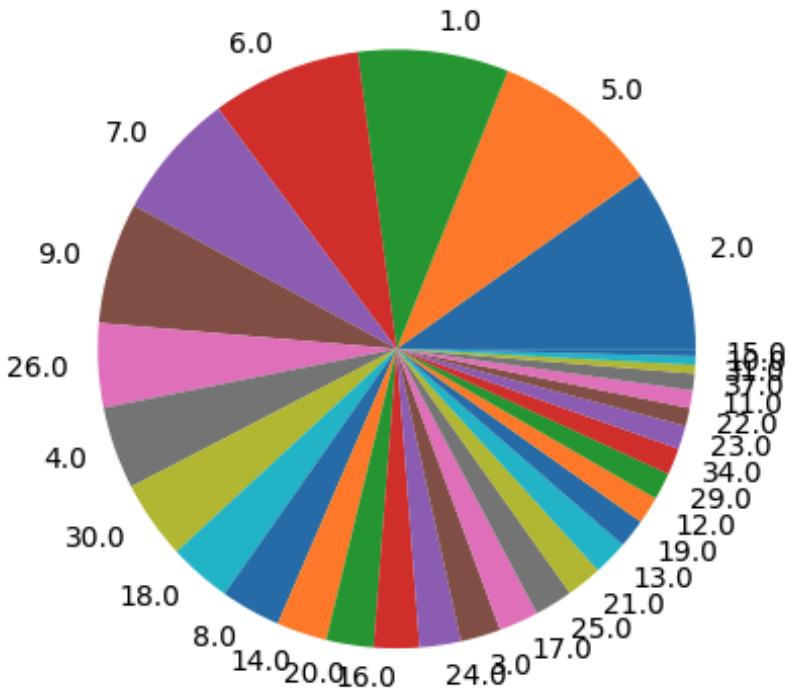

Figure 4: Centers distribution plot

Another interesting statistic was to calculate the number of patients that recieved intravenous thrombolysis.

```
nri_thrombolyseiv
Oui            16423
Non            13666
Ne sait pas       74
Name: count, dtype: int64
```

Figure 5: Thrombolysis distribution statistic

Figure 6: Thrombolysis distribution plot

This next statisctic show the number of patients for each grade of MRS at 3 months (Lower is better).

```
mrs_3mois
mRS 6    6325
mRS 1    4886
mRS 3    3903
mRS 0    3556
mRS 2    3444
mRS 4    2858
mRS 5    1626
Name: count, dtype: int64

Percentages:
 mrs_3mois
mRS 6    23.779983
mRS 1    18.369802
mRS 3    14.674036
mRS 0    13.369426
mRS 2    12.948342
mRS 4    10.745169
mRS 5     6.113242
Name: count, dtype: float64
```

Figure 7: MRS at 3 months distribution statistic

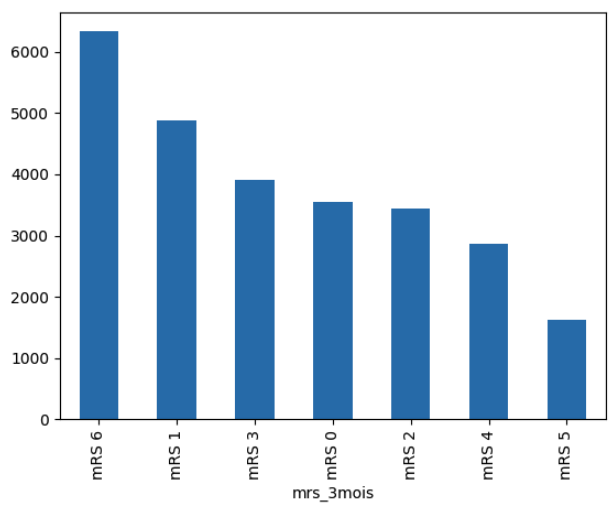

Figure 8: MRS at 3 months distribution plot

One of the most crucial statistic is the aspectj0
(Brain area not affected →10 = perfect, 0 = entire brain affected)

```
aspectj0
8.0     3918
10.0    3859
9.0     3787
7.0     3065
6.0     1871
5.0     1276
4.0      683
3.0      527
2.0      304
1.0      189
0.0      129
Name: count, dtype: int64
Mean = 7.557527539779682
```

Figure 9: AspectJ0 distribution statistic

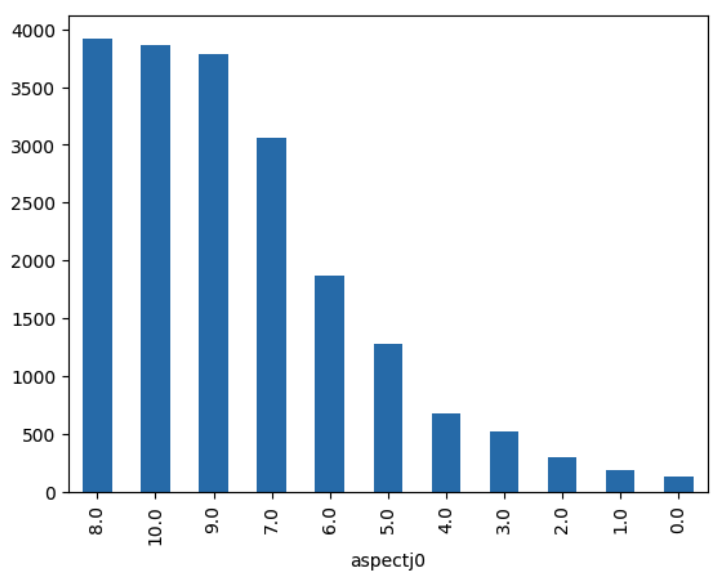

Figure 10: AspectJ0 distribution plot

Another important statistic is the nihssh0 score distribution.
(First clinical evaluation →0= normal 42 = worst)

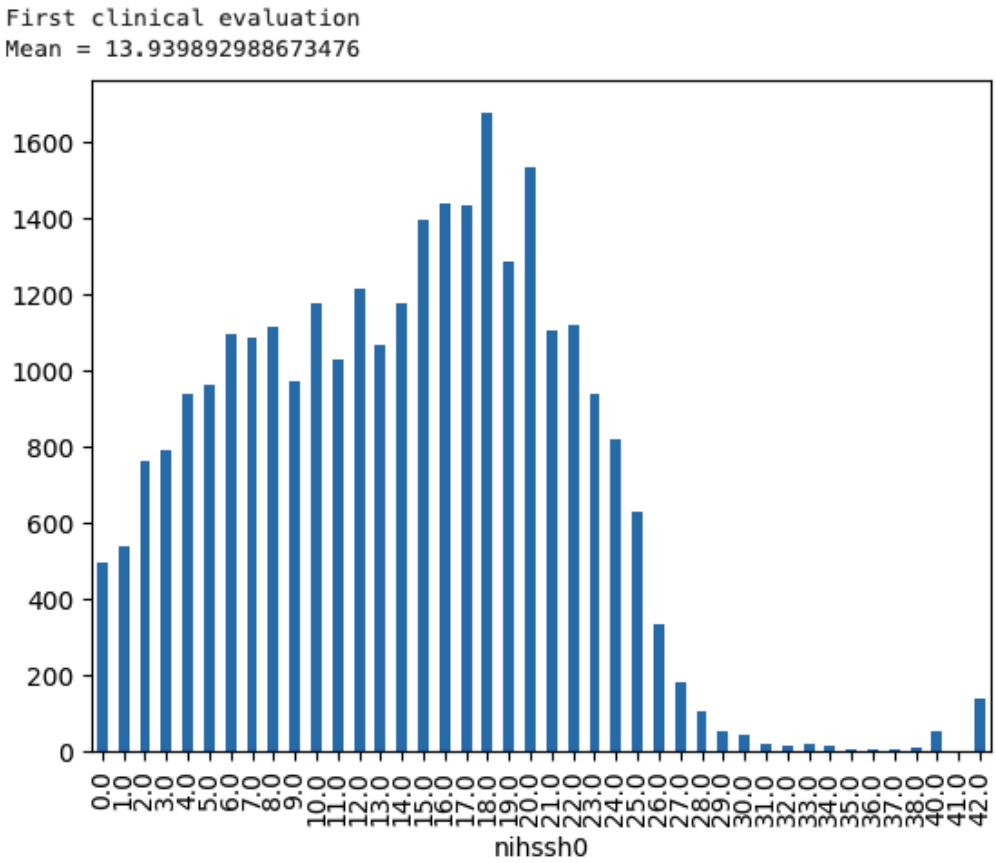

Figure 11: NIHSSH0 distribution Plot and Mean

We have different nihssh scores for different time points, the next one is the nihssh1 score that is a second clinical evaluation (not always available).

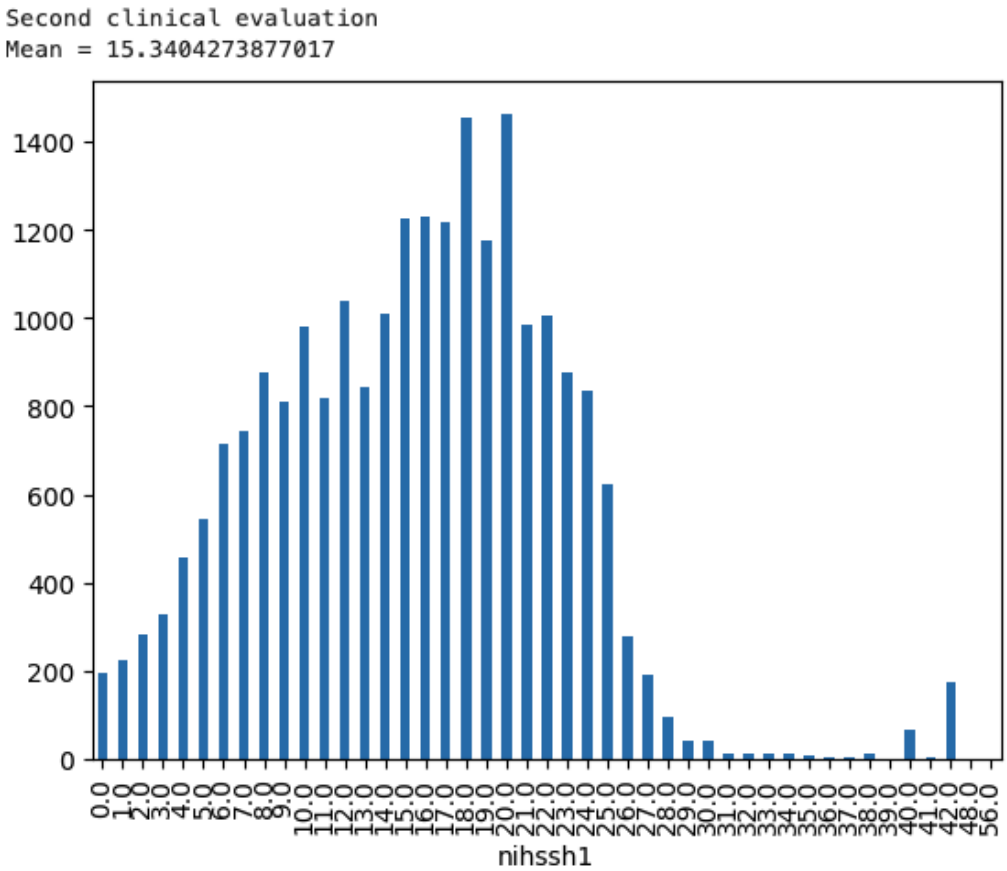

Figure 12: NIHSSH1 distribution Plot and Mean

Here we can see that some scores are off the charts, maybe due to a data entry error, but I will leave them in the dataset for now.

Then we have the nihPost score that is the NIH scale to assess the severity of the neurological deficit

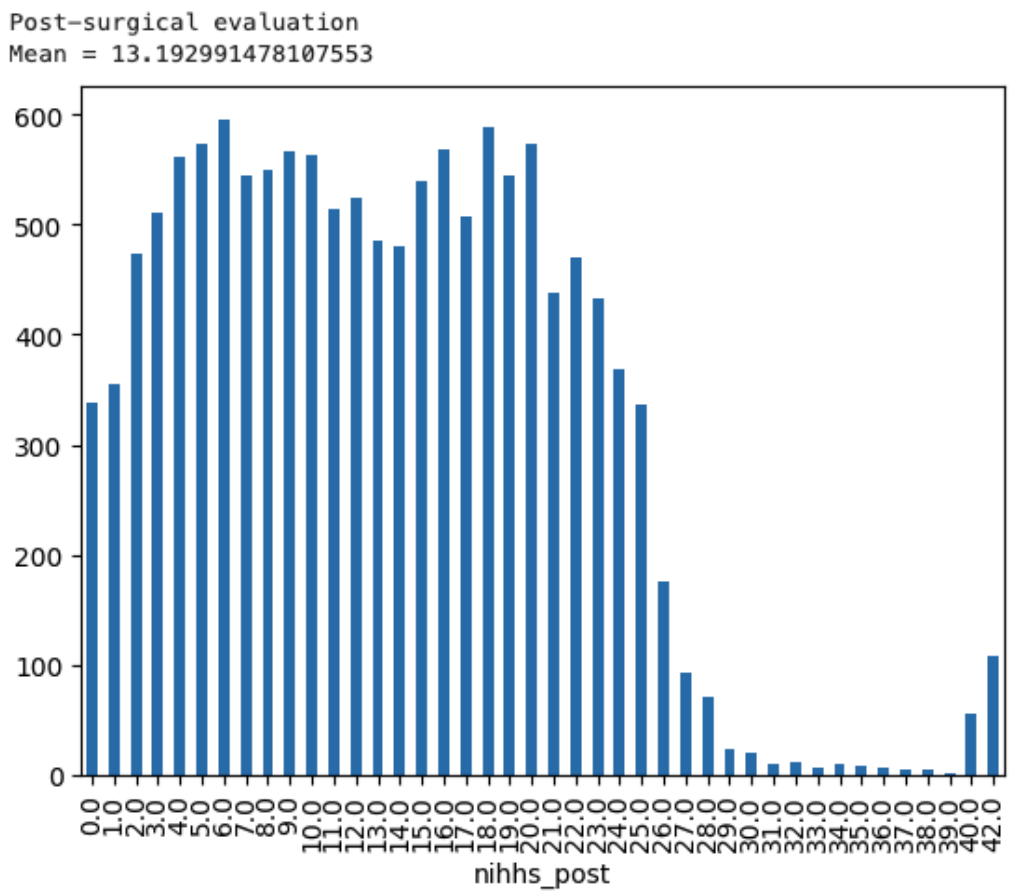

Figure 13: NIHPost distribution Plot and Mean

Finally we have the Nih score at 24 hours, here we can appreciate that the mean is lower than the previous scores and we can also see that there is so much more 0 scores and the distribution has shifted towards the lower scores.